UniPulse: AI-Based Social Integration and Inclusion Platform for University Freshers

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***Abstract*—This research investigates the problem of inadequate social integration, emotional distress, and delayed support among university freshers, which contributes to isolation, poor wellbeing, and increased dropout rates, while existing student support systems remain largely reactive, fragmented, and non-personalized. To address this, the study proposes UniPulse, an AI-driven social integration and inclusion platform that provides proactive, personalized, and scalable student support through four integrated modules: Adaptive Student Profiling and Learning Persona Management, Intelligent Peer Matching and Relationship Evolution Management, Social Inclusion Risk Detection and Early Intervention Management, and Conversational Support and Student Wellbeing Management. Leveraging behavioral analytics, and emotion recognition, UniPulse continuously builds student profiles, detects social inclusion risks, facilitates meaningful peer connections, and delivers real-time conversational and wellbeing interventions within an ethical and culturally adaptive framework. Experimental evaluation and system analysis indicate that UniPulse improves early identification of at-risk students, increases peer engagement, and enhances overall student wellbeing, demonstrating its effectiveness in reducing attrition risk and transforming traditional student support services into proactive, intelligent, and inclusive academic support ecosystems.**

Keywords—early intervention, adaptive profiling, peer matching, conversational support, university freshers

I. Introduction

The transition to university life presents significant academic, social, and emotional challenges for first-year students, often leading to anxiety, depression, social isolation, and increased attrition [1]. Research indicates alarming rates of these issues, with 38% of students experiencing anxiety, 34% reporting depression symptoms [2], and nearly 65% affected by social isolation [3]. In Sri Lanka, these challenges are compounded by cultural stigma, limited access to counseling, and institutional gaps in proactive student support [4]. Traditional support systems, such as manual counseling, static mentoring programs, and generic orientation activities, are often reactive, fragmented, and unable to scale effectively across diverse student populations [5].

Current AI-based solutions in education tend to focus on narrow domains such as academic performance prediction or standalone mental health chatbots, lacking integration with holistic student ecosystems [6]. There remains a significant research and implementation gap in developing culturally contextualized, adaptive, and explainable AI systems that can dynamically address the multifaceted nature of student wellbeing and social integration in higher education.

This research proposes UniPulse, a comprehensive AI-driven platform that unifies four synergistic modules into a single support ecosystem:

1. Adaptive Student Profiling and Learning Persona Management – Continuously models student behavior, motivation, and emotional states to create dynamic learning personas.
2. Intelligent Peer Matching and Relationship Evolution Management – Uses weighted compatibility algorithms to foster meaningful peer connections and adaptively rematch students based on weekly feedback.
3. Social Inclusion Risk Detection and Early Intervention Management – Monitors academic and social engagement to identify at-risk students and trigger timely, personalized interventions.
4. Conversational Support and Student Wellbeing Management – Provides empathetic, AI-driven conversational support and escalates critical cases to human counselors.

UniPulse is designed to be scalable, ethically compliant, and sensitive to the Sri Lankan educational and cultural context. By integrating multimodal student data, including LMS interactions, self-assessments, and social engagement metrics. The platform enables early detection, personalized support, and continuous adaptation to student needs. This research aims not only to develop a functional platform but also to establish a new paradigm for AI-enhanced student support that is proactive, inclusive, and transformative. Figure 1 depicts the full system conceptual diagram.

Fig. 1. Full System Conceptual Diagram.

II. Literiture Review

First-year students frequently experience high rates of anxiety, despair, and loneliness, which highlights the psychological and social difficulties that come with adjusting to university life [1]. These issues are severe in Sri Lanka because of limited access to mental health resources, cultural stigma surrounding help-seeking, and institutional gaps in scalable, personalized support [5] [3]. While traditional interventions such as orientation programs and face-to-face counseling remain valuable, they often lack the capacity to deliver continuous, adaptive, and proactive care across diverse student populations [3].

Artificial intelligence (AI) is becoming more and more important in addressing student retention and well-being, as seen by recent developments in educational technology. AI systems have demonstrated potential in the field of adaptive student profiling by using behavioral and engagement data to tailor learning experiences [6]. However, real-time emotional, motivational, and social aspects that are crucial for comprehensive student support are not integrated into current profiling models, which primarily concentrate on academic indications [7]. With its adaptive profiling module, UniPulse seeks to fill a significant gap in systems that dynamically change with the student, updating profiles every week to reflect changing needs and states.

In order to identify at-risk students, early risk detection methods in higher education have historically focused on academic performance criteria like attendance and grades [8]. However, through sentiment analysis, engagement tracking, and predictive modeling, contemporary learning analytics make it possible to identify non-academic risk factors including mental distress and social isolation [9]. Despite these developments, the majority of systems lack actionable escalation paths that link detection to prompt response and are not tailored to Sri Lanka's institutional and cultural setting. In order to facilitate early, culturally appropriate treatments, UniPulse's risk detection module incorporates behavioral, emotional, and social indicators.

Studies have shown that conversational AI for mental health support is effective in lowering anxiety and offering low-stigma support, making it a scalable and accessible form of intervention [10]. AI-powered chatbots have been incorporated into several student support frameworks and can provide sympathetic, real-time chats [11]. However, these systems are rarely integrated into institutional care routes that enable smooth escalation to human counselors when necessary, and they frequently lack contextual understanding of academic stressors. The conversational support module of UniPulse is built to function within a tiered support model, guaranteeing that high-risk situations are quickly escalated while sustaining ongoing AI-driven engagement.

In order to promote social integration, which is closely associated with student retention and wellbeing, intelligent peer matching has been identified as an essential component [12]. According to research, compatibility-based matching, which takes into account communication styles, personality traits, and academic interests can improve the development of significant peer connections [13]. Nevertheless, few of the current digital matching tools provide mechanisms for dynamic re-matching based on continuous interaction input, and they frequently operate as standalone platforms without integration into larger student support ecosystems [14]. By using weighted scoring algorithms and ongoing connection evaluation, UniPulse's peer matching module aims to get beyond these restrictions.

Even while the use of AI in education is growing, the majority of solutions are still disjointed, focusing on social integration, mental health, or academic achievement separately [6] [7] [10]. An integrated platform that integrates risk detection, intelligent peer matching, adaptive profiling, and conversational support into a coherent, proactive, and morally sound environment is obviously needed. In response to this requirement, UniPulse provides a unified, culturally contextualized AI platform that not only identifies and resolves student issues early on but also promotes a welcoming and inclusive campus community.

III. Materials and Methods

This study proposes the design and development of an AI-based Social Integration and Inclusion Platform named UniPulse, aimed at supporting first-year university students during student's transition into higher education. To detect emotional, motivational, and social risks early and offer tailored solutions and peer support, the platform combines artificial intelligence, data-driven behavioral analysis, and human-centered design concepts.

A. Data Collection

Data for the study are collected from 120 first-year university students, who represent the primary target group of the proposed system. First-year students experience major academic, social, and emotional transitions during the initial year of university life, making participant responses valuable for examining peer compatibility, wellbeing, and engagement. To ensure consistency and reliability, a structured weekly form is administered to participants throughout the study period.

The weekly form is designed to capture information related to emotional states, stress levels, social interactions, academic comfort, and overall wellbeing. A recurring data collection approach enables continuous observation of variations in student experiences over time rather than relying on single-point responses. Weekly monitoring supports the early identification of emerging behavioral patterns, trends, and potential challenges encountered by students.

B. *System Components*

- *Adaptive Student Profiling and Learning Persona Management*

The weekly form is designed to capture information related to emotional states, stress levels, social interactions, academic comfort, and overall wellbeing. A recurring data collection approach enables continuous observation of variations in student experiences over time rather than relying on single-point responses. Weekly monitoring supports the early identification of emerging behavioral patterns, trends, and potential challenges encountered by students. such as orientation programs and face-to-face counseling remain valuable, they often lack the capacity to deliver continuous, adaptive, and proactive care across diverse student populations [3].

The student profiling process begins with an onboarding and profile creation phase, where first-year students self-register by providing essential personal, academic, and well-being-related details. This includes basic demographic information, degree program, living status, initial emotional state, motivation level, and perceived social integration level. Upon successful registration, the system automatically creates an individual student profile, assigns a unique identification number, initializes baseline wellbeing scores, and records the registration timestamp. This initial dataset forms the foundation of the student's adaptive learning persona, which is continuously refined as new information becomes available.

Following profile creation, the system implements a structured weekly check-in workflow to ensure continuous data collection. A weekly form is generated every seven days based on the student's last submission date. When a student logs in, the system compares the current date with the last submission date, if the difference is seven days or more, the weekly check-in form is displayed. If not, the form remains hidden until the next eligible time. The form captures data across three core domains: motivation, social inclusion, and emotional wellbeing. Each domain contains multiple questions, and a scoring mechanism calculates the average score for each domain using the formula: total responses divided by the number of questions. These scores are then used to update the student's profile, with timestamps recorded to support trend analysis. Emotional risk values are fetched from the Social Inclusion Risk Detection module via an integrated service.

The updated data is visualized through a personalized dashboard that displays motivation status, inclusion status, emotional risk level, and overall well-being classification using color-coded indicators (green, amber, and red). A logistic regression model is applied to classify overall wellbeing by analyzing motivation score, social inclusion score, and emotional risk level, producing a probability of risk. Students with a probability greater than 0.6 are categorized as "At Risk," while others are labeled as "Good." Based on this classification, a rule-based recommendation engine generates personalized suggestions, such as peer matching, chatbot support, or motivational prompts. Additionally, the system stores weekly data for the most recent four weeks to perform trend analysis, generating visual graphs for motivation, inclusion, emotional risk, and risk versus peer match effectiveness. This continuous, adaptive approach ensures that student profiles remain accurate, responsive, and supportive throughout the academic journey. Figure 2 shows the conceptual diagram of Adaptive Student Profiling and Learning Persona Management.

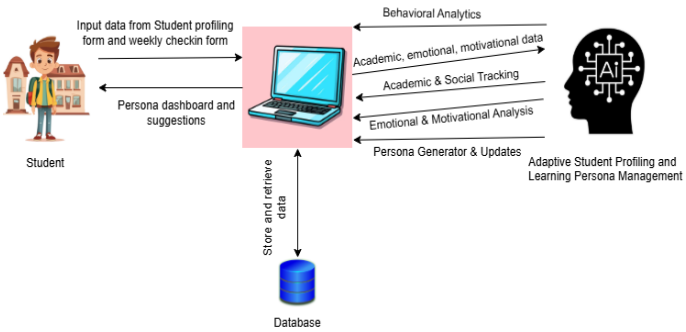

Fig. 2. Adaptive Student Profiling and Learning Persona Management.

- *Social Inclusion Risk Detection and Early Intervention Management*

The Social Inclusion Risk Detection and Early Intervention module collects weekly self-assessment data from first-year students through a short survey. The survey covers five key wellbeing areas: stress and anxiety, depressive mood, disengagement and burnout, social connection and isolation, and openness to seeking help. Each question uses a five-point Likert scale, and selected items for social connection and openness are reverse-scored to ensure consistent interpretation. The system calculates an average score for each domain, creating a standardized emotional profile for each student, while assigning higher scores to more severe conditions for stress, depression, and disengagement.

To ensure reliable analysis, the module applies a four-week weighted scoring system, giving more importance to recent responses; 50% for the current week (W0), 20% each for the previous two weeks (W1 and W2), and 10% for the fourth week(W3). Weighted Score formula is:

$$(W0 \times 0.50) + (W1 \times 0.20) + (W2 \times 0.20) + (W3 \times 0.10)$$

Weighted scores are then classified into risk levels (low, moderate, high), while trend detection determines whether the student's emotional state is improving, stable, worsening, or experiencing a sudden drop. This dual approach captures both the severity of emotional risk and changes over time, providing a more complete picture of each student's wellbeing.

Based on the combined risk level and trend, the system maps the student's condition to a priority category and generates personalized recommendations. Suggestions range from urgent professional support to preventive measures such as short social interactions or simple stress management techniques. When multiple areas show risk, the system prioritizes

recommendations in the order of mood/depression, stress/anxiety, social isolation, disengagement, and openness. This methodology ensures early detection of emotional challenges, timely intervention, and ongoing support to help first-year students maintain student's wellbeing throughout the academic year. Figure 3 represents the conceptual diagram of Social Inclusion Risk Detection and Early Intervention Management.

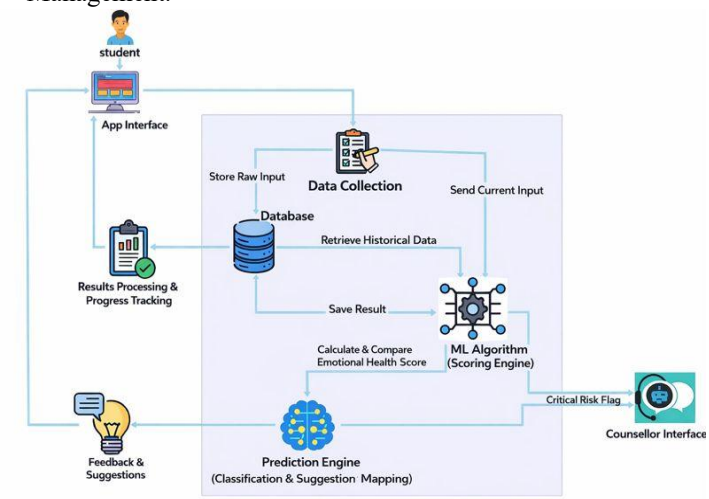

Fig. 3. Social Inclusion Risk Detection and Early Intervention Management.

- *Conversational Support and Student Wellbeing Management*

The Conversational Support and Student Wellbeing Management module operates through an AI-powered chatbot that provides students with continuous emotional assistance, guidance, and wellbeing-related support. Students can interact with the chatbot freely to express concerns, seek advice, or discuss academic and social challenges. The system is integrated with the weekly wellbeing assessment module so that conversations are context-aware, allowing responses to be personalized based on the student's current emotional state, risk level, and historical trends. All interactions are securely logged to enable pattern recognition and continuous improvement of the support process.

To ensure student safety, the chatbot applies a crisis keyword severity scoring mechanism that classifies messages into three levels: Red Flag (Critical), Yellow Flag (Concerning), and Blue Flag (Warning). This is achieved through real-time keyword and phrase detection. Critical phrases related to self-harm or suicidal ideation trigger immediate automated escalation, activating emergency protocols and sharing crisis support resources. Concerning keywords such as expressions of hopelessness or emotional emptiness activate empathetic de-escalation dialogues and flag the case for human counselor review. Warning-level phrases such as feeling overwhelmed, anxious, or loneliness initiate increased emotional support, coping strategies, and more frequent check-ins. This layered response structure ensures that the system reacts proportionately and responsibly to the severity of emotional distress.

Finally, the chatbot prioritizes its guidance based on urgency and emotional risk level. High-severity interactions focus on safety, emotional grounding, and referral to human support, while moderate and low-risk interactions emphasize coping strategies, motivation, reflection, and social engagement. The system avoids overwhelming students by providing only the most relevant guidance at a time. By combining conversational AI, emotional trend data, and crisis-level keyword detection, this module ensures that students receive timely, empathetic, and appropriate wellbeing support, reinforcing emotional resilience and early intervention throughout their academic journey. Figure 4 depicts the conceptual diagram of Conversational Support and Student Wellbeing Management.

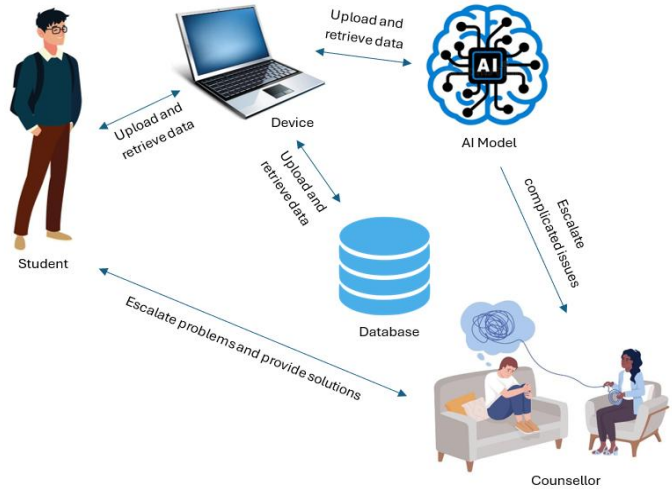

Fig. 4. Conversational Support and Student Wellbeing Management.

- *Intelligent Peer Matching and Relationship Evaluation Management*

The Intelligent Peer Matching and Relationship Evaluation Management component is designed to create personalized, compatible peer connections for first-year university students. The process begins with students completing a detailed survey covering personality traits, academic background, hobbies, lifestyle, and wellbeing. The system then calculates a weighted compatibility score for each student pair, considering multiple dimensions such as interests, academics, personality, emotional wellbeing, communication style, and social preferences. This ensures that suggested peers are well-matched across both social and academic contexts, promoting balanced and supportive interactions.

The system assigns specific weights to each category to prioritize factors that influence effective peer matching. Interest match and hobbies (25%) assess shared passions, learning styles, and group activity participation, while academic compatibility (20%) aligns students with peers from similar faculties or streams. Personality compatibility (20%) considers introversion-extroversion levels and comfort in groups, and emotional and wellbeing alignment (15%) uses weekly check-ins to capture mood, stress, and social integration levels. Additionally, communication preferences (10%) and social preferences (10%) ensure students are paired with peers who share compatible interaction styles and social settings. These weighted scores are combined to generate a ranked list of suggested peers.

Beyond one-to-one matches, the platform supports dynamic peer interactions and group formations. Students can send

connection requests, initiate chats, or create and discover study or support groups. Weekly updates on stress, mood, and social engagement allow the system to adjust compatibility scores in real time, ensuring that peer suggestions remain relevant as relationships evolve. By combining data-driven compatibility calculations with ongoing monitoring, the component facilitates meaningful peer connections, supports social inclusion, and helps students adapt to university life successfully. Figure 5 shows the conceptual diagram of Intelligent Peer Matching and Relationship Evaluation Management.

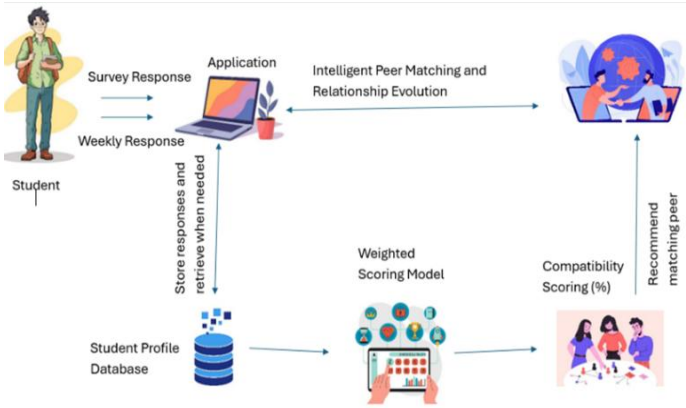

Fig. 5. Intelligent Peer Matching and Relationship Evaluation Management.

C. *User Testing and Refinements*

The complete UniPulse platform, including student profiling, risk detection, conversational support, and intelligent peer matching, was tested by first-year students to evaluate functionality, usability, and effectiveness. Students actively used the weekly check-ins, received recommendations, engaged in peer matching, and participated in chats and group activities. The feedback was overwhelmingly positive, highlighting the system's ease of use, helpfulness of personalized suggestions, and effectiveness in improving social integration and emotional support. Insights from testing were incorporated to fine-tune the interface, recommendation logic, and matching algorithms, ensuring the platform delivers a reliable, engaging, and supportive experience for all users.

IV. Results and Discussion

A. *Objective Achivements*

The proposed AI-based social integration platform was successfully developed and implemented to address the challenges of peer isolation, emotional disengagement, and lack of adaptive support among university students. The system achieved its primary objectives through the integration of intelligent profiling, peer matching, conversational support, and emotional risk detection mechanisms.

- *Adaptive Profiling Effectiveness:*

The system demonstrated the ability to generate continuously evolving student profiles by integrating academic, behavioral, emotional, and social interaction data. Unlike static profiling approaches, adaptive profiling accurately reflected changes in student wellbeing over time, enabling personalized system responses through dashboard as shown in Figure 6 below.

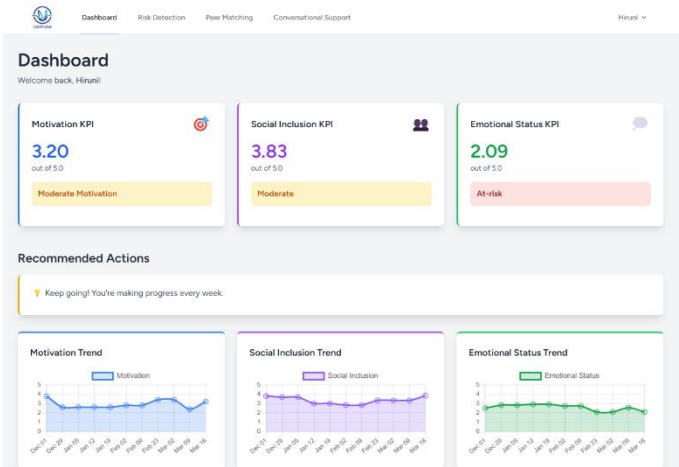

Fig. 6. Personalized system responses and wellbeing trends.

- *Peer Matching Accuracy:*

The intelligent peer matching module produced multidimensional compatibility scores by combining personality traits, interests, emotional indicators, and communication preferences. Matches generated through this approach resulted in higher engagement continuity compared to conventional self-selected or academically grouped peer formations. Common interest between peers can be seen as shown in figure 7.

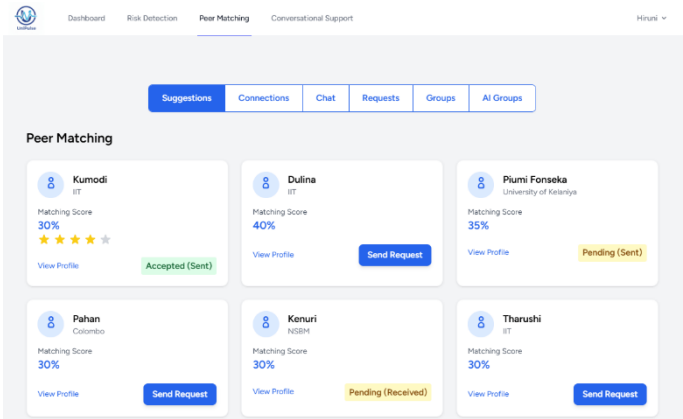

Fig. 7. Common interest between peers.

- *Conversational Support Responsiveness:*

The conversational flow mechanism successfully guided students through structured yet empathetic interactions. Context-aware prompts encouraged regular check-ins, self-reflection, and peer interaction without creating pressure or dependency, supporting consistent platform engagement. Figure 8 shows the empathetic support that the conversational support provides.

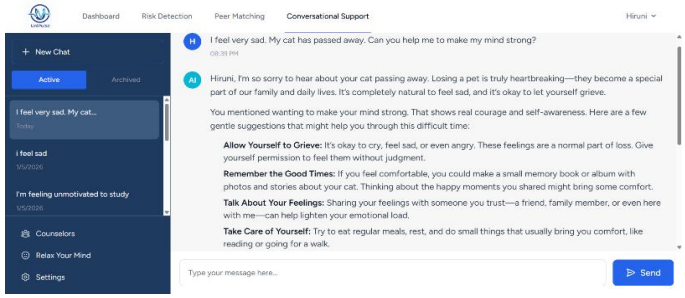

Fig. 8. Empathetic conversational support.

- *Emotional Risk Detection Reliability:*

The emotional risk detection component effectively identified early signs of emotional distress, social withdrawal, and disengagement by analyzing longitudinal behavioral and emotional trends. The system enabled early-stage interventions before issues escalated into critical academic or psychological outcomes. Figure 9 illustrates the depression risk analysis dashboard, highlighting weekly emotional trend monitoring and risk level classification used to support early detection.

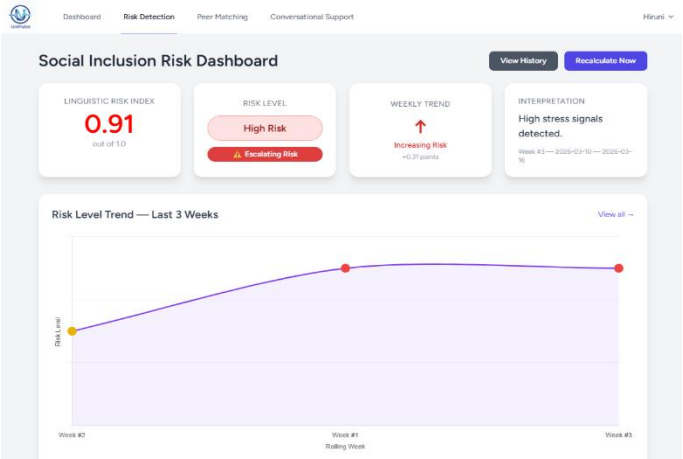

Fig. 9. Weekly depression risk trend detection.

B. *Bridging the Research Void*

Prior research and existing university support systems largely focus on static onboarding programs, academic mentoring, or reactive counseling services. The outcomes of this study directly address these gaps by introducing a proactive, adaptive, and AI-driven social integration framework.

- Dynamic and Continuous Support: Unlike traditional one-time interventions, the proposed system continuously adapts to student needs through real-time profile updates and evolving peer recommendations. This ensures sustained relevance throughout the student lifecycle.
- Integration of Emotional Intelligence: Existing peer matching platforms rarely incorporate emotional wellbeing as a core parameter. By embedding emotional indicators and conversational feedback into the system, the platform bridges the gap between social networking tools and student mental health support systems.
- Reduction of Social Inequality: The platform minimizes reliance on informal social networks, which often disadvantage introverted, international, or marginalized students. Algorithmic peer recommendations promote inclusivity by ensuring equitable access to meaningful social connections.

CONCLUSION

UniPulse demonstrates that an AI-powered platform can effectively address the critical challenges of social isolation and emotional distress experienced by university freshers. By integrating adaptive student profiling, intelligent peer matching, risk detection, and conversational support into a single ecosystem, the system moves beyond reactive, fragmented support models. The positive testing outcomes and the accuracy of its core modules confirm that UniPulse successfully enables early identification of at-risk students, fosters meaningful peer connections, and delivers timely, personalized interventions.

Ultimately, this research establishes a scalable, culturally adaptive model for technology-enhanced student wellbeing that promotes inclusivity while preserving ethical safeguards and human oversight.

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